ME 5309 – Mid-Semester Examination: ANSWER SHEET

Roll No.: ME25M048 **Name:** Rajat Mukherjee

Questions attempted: Q4

Question 4

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\, r$, i.e. $s^2 + 4s + 8 = 0$.

$s = [-4 \pm \sqrt{(16 - 32)}]/2 = -4/2 \pm \sqrt{(-16)} = \mathbf{-2 \pm 4j}$.

DC gain $= N_{ff}/8 = 1$, so $\mathbf{N_{ff} = 8}$.

(b) With $b = 0.75$ ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

$\theta_{ss} = \mathbf{0.8}$. Tracking error $= (1 - 0.8)/1 \times 100 = \mathbf{20\%}$. (Less than the 25% actuator loss, because the stiffness also dropped.)

(c) Substituting u and differentiating: the integral term enters with a minus sign, so the characterstic polynomial is $s^3 + (a + bk_2)s^2 + (c + bk_1)s - bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

So $2 + k_2 = 8 \Rightarrow k_2 = 6$; $6 + k_1 = 24 \Rightarrow k_1 = 18$; $-K_I = 32 \Rightarrow \mathbf{K_I = -32}$.

(d) Because the closed loop is stable, the steady-state error is zero for any b and c.

Failure mode: integrator windup. Remedy: clamp the integrator output (anti-windup).